

AYUSHI MAITY

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PROFESSIONAL SUMMARY

Research and clinical professional with hands-on experience spanning preclinical in vivo pharmacology, CAP/CLIA-regulated diagnostic assay development, GMP quality control, and direct patient care. GCP- and Human Subjects Protection-certified, with a consistent record of building and validating new methods, authoring regulatory-ready documentation, and translating technical findings for physicians, regulatory reviewers, and cross-functional teams.

EDUCATION

MS in Medical Sciences | Boston University School of Medicine Sep 2024 – May 2026

BS in Biochemistry, Certificate in Leadership | Rider University Sep 2018 – May 2022

Honors: Endowed Chemistry Research Fellowship, Wyeth Ayerst Achievement Award, Dean's List, Dean's Scholarship

CERTIFICATIONS

CITI Good Clinical Practice (GCP) — ICH E6 GCP Investigator Site Training | Oct 2025 – Oct 2028

CITI Basic Human Subjects Protection — Stage 1 Basic Course | Oct 2025 – Oct 2028

CITI Institutional Animal Care and Use Committee (IACUC) Certification

CITI Working with Mice in Research Certification

PUBLICATIONS

Maity, A. (2026). FGF21 Agonism: Mode of Action and Therapeutic Efficacy to Prevent Cirrhosis and Its Sequelae in Mouse Models of End-Stage Liver Disease. ProQuest Dissertations & Theses Global.

PROFESSIONAL EXPERIENCE

Medical Assistant Jul 2026 – Present

Windsor Medicine (MDVIP Concierge Internal Medicine Practice)

- Coordinate clinical care delivery for 30+ patients weekly across 2 physicians, managing intake, chief-complaint documentation, vitals, and informed consent discussions for in-office procedures.
- Perform phlebotomy, urinalysis, and EKG testing at every patient encounter, enabling same-visit diagnostic decision-making.
- Deliver MDVIP preventive-care screening through spirometry, pulmonary function test, ankle-brachial index, hearing/vision — per standardized protocol, sustaining consistent, high-quality wellness exams.
- Maintain audit-ready clinical documentation in eClinicalWorks EHR across every patient visit.

Gastroenterology Research Intern Sep 2025 – May 2026

Beth Israel Deaconess Medical Center – Harvard Medical School

- Executed a preclinical in vivo evaluation of an FGF21 metabolic agonist for cirrhosis prevention under PI Dr. Yury Popov, successfully completing 3 independent longitudinal studies across 200+ mice to accelerate dosing and efficacy assessment for the compound.

Optimized and independently managed a breeding mouse colony, implementing systematic tracking of litters and ear-notching genotypes to sustain a consistent, accurately documented animal supply for ongoing laboratory studies.

- Enforced strict protocol compliance during compound administration (subcutaneous/intraperitoneal) and routine sampling across every longitudinal timepoint, per IACUC-approved procedures.
- Performed pre-, peri-, and post-operative surgical procedures including cardiac perfusion, liver isolation surgery, and tumor removal, followed by humane euthanasia, necropsy, and terminal tissue collection following IACUC guidelines, generating samples for downstream analysis.
- Quantified hepatic fibrosis using microtomy, H&E/Sirius Red staining, microscopy, and hydroxyproline assays, providing a key dataset that directly informed dosing and efficacy assessment for the compound.
- Maintained audit-ready experimental records and calibrated laboratory equipment in strict compliance with institutional SOPs, successfully ensuring the laboratory's constant state of readiness for regulatory review.

Microbiologist Mar 2024 – Aug 2024

Kenvue (formerly Johnson & Johnson Consumer Health)

- Safeguarded regulatory-submission readiness across 6+ topical and ingested/OTC personal care and consumer health product lines — including Neutrogena sunscreens, Aveeno Baby products, Bengay, concealers, and bronzers — by executing GMP-compliant microbiological testing that fed directly into R&D pipeline decisions and regulatory submissions.
- Performed a broad range of microbiological test methods — MLT, MCT, Growth Promotion Testing, TMS, water activity testing, Gram staining, and microorganism identification to ensure accurate product quality assessments across every test cycle.
- Cultured and maintained 7-8 microbial strains weekly under strict aseptic technique for preservative efficacy/Antimicrobial Effectiveness Testing (AET) in lotions, creams, and liquids, achieving 100% audit-ready datasets.
- Reviewed 100+ laboratory data entries in LIMS for data integrity and cGMP compliance, investigating out-of-specification results and identifying root causes that informed corrective actions.
- Prevented testing schedule disruptions across concurrent product lines by managing lab inventory.

Research Associate

Jun 2022 – Jul 2023

Genesis Global Group - Institute for Biomarker Research (IBR)

- Delivered 3 CAP/CLIA-accreditation-ready diagnostic assays across 300+ clinical specimens — including a multiplex panel detecting 3 pathogens (BV, yeast, Trichomoniasis) from a single swab — by isolating DNA and validating PCR/RT-qPCR methods from the ground up.

Established clinical-grade assay accuracy for patient reporting by engineering a two-tier Mycoplasma genitalium testing algorithm — PCR screening followed by Sanger sequencing confirmation on positive results.

- Automated large-scale PCR processing on a Hamilton Vantage liquid handler and automated plate readers, increasing throughput across high-volume specimen batches.
- Tracked and reconciled 300+ specimens with full chain-of-custody integrity from intake through result reporting, supporting audit-ready laboratory operations.
- Authored SOPs and validation reports underpinning successful CAP/CLIA accreditation, and partnered with physicians and regulatory teams on statistical analysis and manuscript development.

Biochemistry Lab Research Assistant

Jun 2021 – May 2022

Rider University

- Engineered a novel fluorescence-based kinetic assay (mBBr dye) to solve the field's lack of a convenient method for measuring BHMT activity, a liver enzyme linked to cardiovascular disease risk.
- Validated assay sensitivity and optimized buffer/pH conditions via UV-Vis/fluorescence spectrophotometry, identifying CAPS pH 10 as the most sensitive, reproducible condition for BHMT kinetics.
- Established a reusable platform for future BHMT inhibitor screening by presenting validated findings at the Rider University ISCAP research symposium and contributing to manuscript preparation.

ADDITIONAL EXPERIENCE

Organic Chemistry Lab Assistant

Sep 2021 – Dec 2021

Rider University

- Mentored 20 students in organic chemistry lab technique and safety, strengthening independent lab competency across the semester.

Dental Office Assistant

Dec 2021 – Jun 2022

Hometown Orthodontic Care

- Streamlined patient flow and EMR documentation for 20+ patients daily in a high-volume orthodontic practice.

Hospital Intern

Jan 2020

Capital Health

- Gained early exposure to informed consent processes and patient eligibility/enrollment considerations across 8 hospital departments (OB/GYN, Radiology, Gastroenterology, ER, surgical unit, EMT, wound care, pediatrics) under 6+ physicians in this one-month internship.